Lecture 2: Abstract Algebra for Computer Sciences

MATHEMATICS DEPARTMENT

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I. Implementation of Linear Algebra and Abstract Algebra in CS: Hill Cipher

- **Hill Cipher is a symmetric block cipher technique developed by the mathematician Lester Hill in 1929.**
- **Based on Linear Algebra and Abstract Algebra.**
- **Each letter is treated as a digit in base 26:**

A	B	c	d	E	f	g	H	i	j	k	l	m
0	1	2	3	4	5	6	7	8	9	10	11	12

N	O	p	q	R	s	t	U	v	w	x	y	z
13	14	15	16	17	18	19	20	21	22	23	24	25

- **The plaintext is divided into blocks of n letters (considered as a vector of n dimensions) and multiplied by the key (n×n matrix) modulo 26.**
- **The components of the matrix are the key and should be random, provided that the matrix is invertible to ensure decryption is possible.**
- **If the block size is (2×1), then the key matrix should be of size 2×2, and the encryption process will produce ciphertext with two numerical values.**

➢ Encryption

C = KP mod 26, where

P: plaintext

C: Ciphertext

K: The key.

If P is of size n×1, then the key K must be of size n×n.

$$C = KP \bmod 26$$

$$= \begin{bmatrix} k_{11} & k_{12} \\ k_{21} & k_{22} \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \end{bmatrix} \bmod 26 = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

Example: **Encrypt the message "gold" by using Hill Cipher with the key**

$$K = \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix}$$

Solution: **Divide the message into vectors:** $\begin{pmatrix} g \\ o \end{pmatrix} = \begin{pmatrix} 6 \\ 14 \end{pmatrix}$ **and** $\begin{pmatrix} l \\ d \end{pmatrix} = \begin{pmatrix} 11 \\ 3 \end{pmatrix}$

$$C = KP \bmod 26 = \begin{pmatrix} 2 & 3 \\ 3 & 6 \end{pmatrix} \begin{pmatrix} 6 \\ 14 \end{pmatrix} = \begin{pmatrix} 2 \\ 24 \end{pmatrix} \rightarrow \begin{pmatrix} c \\ y \end{pmatrix}$$

$$C = KP \bmod 26 = \begin{pmatrix} 2 & 3 \\ 3 & 6 \end{pmatrix} \begin{pmatrix} 11 \\ 3 \end{pmatrix} = \begin{pmatrix} 5 \\ 25 \end{pmatrix} \rightarrow \begin{pmatrix} f \\ z \end{pmatrix}$$

Then "gold" is encrypted to "cyfz".

➢ **Decryption**

$$C = KP \bmod 26 \Rightarrow P = K^{-1}C \bmod 26$$

Thus, we can decrypt if and only if K^{-1} exists $(gcd(det(K), 26) = 1)$.

Example: Decrypt the cipher "cyfz" by using Hill Cipher with the key

$$K = \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix}$$

Solution:

$$C = K^{-1}P \bmod 26.$$

What is K^{-1}?

$$|K| = 12 - 9 = 3, \quad gcd(3,26) = 1.$$

Then K is invertible.

$$K^{-1} = \frac{1}{|K|} K = \begin{bmatrix} 6 & -3 \\ -3 & 2 \end{bmatrix} \bmod 26 = 3^{-1} \begin{bmatrix} 6 & -3 \\ -3 & 2 \end{bmatrix} \bmod 26$$

What is 3^{-1}?

$$3(9) \bmod 26 = 1 \Rightarrow 3^{-1} = 9$$

$$K^{-1} = 9 \begin{bmatrix} 6 & -3 \\ -3 & 2 \end{bmatrix} \bmod 26 = \begin{bmatrix} 2 & 25 \\ 25 & 18 \end{bmatrix}$$

Divide the message into vectors: $\begin{pmatrix} c \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ 24 \end{pmatrix}$ *and* $\begin{pmatrix} f \\ z \end{pmatrix} = \begin{pmatrix} 5 \\ 25 \end{pmatrix}$.

$$P = K^{-1}C \bmod 26 = \begin{pmatrix} 2 & 25 \\ 25 & 18 \end{pmatrix} \begin{pmatrix} 2 \\ 24 \end{pmatrix} = \begin{pmatrix} 6 \\ 14 \end{pmatrix} \rightarrow \begin{pmatrix} g \\ o \end{pmatrix}$$

$$P = K^{-1}C \bmod 26 = \begin{pmatrix} 2 & 25 \\ 25 & 18 \end{pmatrix} \begin{pmatrix} 5 \\ 25 \end{pmatrix} = \begin{pmatrix} 11 \\ 3 \end{pmatrix} \rightarrow \begin{pmatrix} l \\ d \end{pmatrix}$$

Then "cyfz" is decrypted to "gold".

II. Order of Groups and Order of Elements in Groups:

Definition: **A group $(G, *)$ is called a finite group if G has only a finite number of elements. The order, written $|G|$, of a group $(G, *)$ is the number of elements of G.**

Definition: **A group with an infinite number of elements is called an infinite group.**

Examples:

1. **$(\mathbb{Z}_5, +)$ is a finite group of order 5.**
2. **$\mathbb{Z}, \mathbb{Q}, \mathbb{R}$ are infinite groups with the usual addition.**

Notes:

- **Let $(G, *)$ be a group, $a \in G$, and $n \in \mathbb{Z}$. We now define the integral power a^n of a as follows:**

$$a^0 = e \quad and \quad a^n = \underbrace{a * a * \dots * a}_{n-times}$$

- **Let G be a finite group and $a \in G$.**

  Now $a^2 = a * a \in G$, and by induction, we can show that $a^m \in G$ for all $m \geq 1$.

- **Thus, $\{a, a^2, \dots, a^m, \dots\} \subseteq G$.**

- **Because G is finite, all elements of the set $\{a, a^2, \dots, a^m, \dots\}$ cannot be distinct.**

- **Hence, $a^k = a^l$ for some positive integers $k, l, k > l$.**
- **This implies that $a^{k-l} = e$. Let us write $n = k - l$.**
- **Therefore, $a^n = e$ for some positive integer n.**
- **Also, if G is an infinite group and $a \in G$, then it may still be possible that $a^n = e$ for some positive integer n. This leads us to the following definition.**

Definition: "Order of an element of G"

Let (G, ∗) be a group and $a \in G$. The smallest positive integer n such that $a^n = e$ is called the order of a (denoted by o(a)).
If no such positive integer n exists, then we say that a is of infinite order.

Examples:

1. **In $(\mathbb{Z}_6, +)$, find $o([3])$.**

$$[3]^1 = [3], \quad [3]^2 = [0] = e,$$
$$thus\ o([3]) = 2.$$

2. **In $(\mathbb{Z}_5 *, \times)$, find $o([2])$.**

$$[2]^1 = [2], \quad [2]^2 = [4], \quad [2]^4 = [1] = e,$$
$$thus\ o([2]) = 4.$$

3. **In $(GL(2, \mathbb{R}), \times)$, find $o\left(\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}\right)$ and $o\left(\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\right)$.**

$$\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}^2 = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = e$$

$$thus\ o\left(\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}\right) = 2.$$

And,

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}^2 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}^3 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}^4 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix}$$

By induction we can show that

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}^n = \begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$$

This implies that $o\left(\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\right)$ **is infinite.**

Theorem: Let (G, ∗) be a group and a be an element of G such that o(a) = n.

1. If $a^m = e$ for some positive integer m, then n divides m.
2. For every positive integer t, $o(a^t) = \frac{n}{gcd(t,n)}$

Proof:

1. Since $m, n \in \mathbb{Z}$, by the division algorithm, there exist $q, r \in \mathbb{Z}$ such that

$$m = nq + r,\ where\ 0 \leq r < n.$$

Now,

$$a^r = a^{m-nt} = a^m * a^{-nt} = a^m * (a^n)^{-t} = e * (e)^{-t} = e.$$

But $o(a) = n$. Therefore, n is the smallest positive integer such that $a^n = e$. However, $a^r = e$ and $0 \leq r < n$.

Thus, we must have $r = 0$.

This implies that $m = nq$. Hence, n divides m.

2. Let $o(a^t) = k$, then $a^{kt} = e$, thus by (1), $n \mid kt$.

This implies that $\exists\, r \in \mathbb{Z}: kt = nr$.

Let

$$gcd(t,n) = d, \qquad \exists\, u,v:\ t = du, n = dv, and\ gcd(u,v) = 1.$$

Now,

$$kt = nr \Rightarrow kdu = dvr$$

$$\Rightarrow ku = vr \Rightarrow v \mid ku,$$

and we have

$$gcd(u,v) = 1 \Rightarrow v \mid k,$$

since we have $v = n/d, thus \left(\frac{n}{d}\right) \mid k$.

Now we go to prove that $k \mid \left(\frac{n}{d}\right)$,

$$(a^t)^{\frac{n}{d}} = a^{\frac{nt}{d}} = a^{\frac{ndu}{d}} = a^{nu} = (a^n)^u = e^u = e.$$

Thus by (1), $k \mid \left(\frac{n}{d}\right)$.

As k and n/d are positive integers and each divides the other, then $k = \frac{n}{d}$.

Therefore,

$$o(a^t) = k = \frac{n}{d} = \frac{n}{gcd(t,n)}.$$

III. Subgroups

Definition: Let (G, ∗) be a group and H be a nonempty subset of G. Then (H, ∗) is called a subgroup of (G, ∗) if (H, ∗) is a group, and it is denoted by $(H,*) \leq (G,*)$, simply $H \leq G$.

Theorem: Let G be a group and H be a nonempty subset of G. Then H is a subgroup of G if and only if for all $a, b \in H, ab^{-1} \in H$.

Proof: (for reading)

Suppose H is a subgroup of G. Let $a, b \in H$. Because H is a subgroup, it is a group. Therefore, $b \in H$ implies that $b^{-1} \in H$. Thus, $ab^{-1} \in H$ because H is closed under the binary operation.

Conversely, suppose H is a nonempty subset of G such that $a, b \in H$ implies $ab^{-1} \in H$.

Because $H \neq \emptyset$, there exists $x \in H$.

Let $a = x$ and $b = x$, then

$$e = xx^{-1} \in H, \text{ i.e., H contains the identity.}$$

Next, let $x \in H. Then\ e, x \in H$, implies that

$$x^{-1} = ex^{-1} \in H.$$

Thus, every element of H has an inverse in H. To show that H is closed under the binary operation,

$let\ x, y \in H. Then\ x, y^{-1} \in H$. Thus,

$$xy = x(y^{-1})^{-1} \in H.$$

Hence, H is closed under the binary operation.

Finally,

$$\forall\, x, y, z \in H \subseteq G, thus\ x(yz) = (xy)z.$$

Hence, H is a group, so H is a subgroup of G.

Examples:

1. **Trivial subgroups of a group G are {e} and G itself.**
2. **$(2\mathbb{Z}, +) \leq (\mathbb{Z}, +)$.**

Indeed, let a, b ∈ 2ℤ, then

$$a = 2s, \qquad b = 2t;\ s, t \in \mathbb{Z}$$

Thus,

$$a + b^{-1} = 2s + (-2t) = 2(s - t) \in 2\mathbb{Z}$$

Therefore, $2\mathbb{Z} \leq \mathbb{Z}$.

Corollary: **Let G be a group and H be a finite nonempty subset of G. Then H is a subgroup of G if and only if for all a, b ∈ H, ab ∈ H.**